■ 2.2.3 Indexed Variables

One simple way to set up an "array" in *Mathematica* is to create an explicit list of elements. You can define a list, say `a = {x, y, z}`, then access the elements using `a[[i]]`. One problem with this approach is that you have to fill in all the elements of the list when you first create it. You cannot just fill in the elements you need at a particular time.

You can use transformation rules to set up objects like arrays in a somewhat more flexible fashion.

The idea is simply to define values for expressions like `a[i]`. You can fill in these values when you want to, in any order.

This defines a value for `a[1]`.

```
In[1]:= a[1] = 9
Out[1]= 9
```

This defines a value for `a[2]`.

```
In[2]:= a[2] = 7
Out[2]= 7
```

This shows all the values you have defined for expressions associated with `a` so far.

```
In[3]:= ?a
a
a/: a[1] = 9
a/: a[2] = 7
```

You can define a value for `a[5]`, even though you have not yet given values to `a[3]` and `a[4]`.

```
In[3]:= a[5] = 0
Out[3]= 0
```

Defining values for `a[i]` does not affect the value of `a` itself.

```
In[4]:= a
Out[4]= a
```

This generates an explicit list of the values of the `a[i]`.

```
In[5]:= Table[a[i], {i, 5}]
Out[5]= {9, 7, a[3], a[4], 0}
```

Here is another way to make the same list.

```
In[6]:= Array[a, 5]
Out[6]= {9, 7, a[3], a[4], 0}
```

You can think of `a[i]` as being like an "indexed" or "subscripted" variable. (Section 2.6.7 describes how to make i actually print as a subscript.)

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`a[i] = value`	add or overwrite a value
`a[i]`	access a value
`a[i] =.`	remove a value
`?a`	show all defined values
`Clear[a]`	clear all defined values
`Table[a[i], {i, 1, n}]` or `Array[a, n]`	convert to an explicit `List`

Manipulating indexed variables.

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